

Module 03 Assignment

Software Testing Life Cycle (STLC) with AI

Q1. Create a test plan for online shopping website

Q2. Prepare a Test Checklist

Create a checklist containing 20 validation points for the below features

- Login Checklist
- Registration Checklist
- Checkout Checklist

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PART 1: TEST PLAN DOCUMENT - SAUCEDEMO

1. Test Plan ID and Version

Item	Details
Test Plan ID	TP-SD-001
Version	1.0
Application	SauceDemo Online Shopping Website
Website URL	https://www.saucedemo.com/
Prepared By	Srabony Islam
Testing Type	Functional, UI, Usability & Regression Testing
Testing Level	System Testing

2. Objective

The primary objective of this test plan is to verify that the SauceDemo online shopping application works according to its expected functional requirements and user workflows. The testing will verify that users can:

- Log in successfully using valid credentials and receive appropriate validation for invalid attempts.
- View available products, sort products using all available criteria, and view detailed product specs.
- Add to and remove products from the shopping cart, maintaining exact cart quantities and calculations.
- Proceed through end-to-end checkout by entering customer information and reviewing order overviews.
- Successfully place orders, observe confirmation messages, and log out securely.
- Identify defects affecting system functionality, UI consistency, usability, data integrity, and end-user experience.

3. Scope

3.1 In Scope

The following modules and core customer journeys are strictly within the testing scope:

- Login Module: Valid/invalid login credentials, error messages, and logout functionality.
- Product / Inventory Module: Listing display, product names/descriptions/prices/images, product details navigation, sorting, and add/remove buttons.
- Shopping Cart Module: Cart view, product count badge validation, price aggregation, and item removal.
- Checkout Module: Navigation flow, customer data input validation (First Name, Last Name, Zip Code), and mandatory field enforcement.
- Checkout Overview & Order Completion: Order summary review, tax and total price calculations, order submission, and confirmation screen.
- Navigation & UI Consistency: Cross-page link integrity, header/footer menu operations, and interactive response checks.

3.2 Out of Scope

The following areas are explicitly excluded from this test execution cycle:

- Real payment gateway processing and bank/card transactions.
- Production customer data setup and live logistics/shipping integration.
- Performance, Load, and Stress testing under high concurrency.
- Security penetration testing, vulnerability scanning, and ethical hacking.
- Database-level verification (Direct SQL / NoSQL operations) and backend API testing.
- Mobile Native Application testing (Testing is restricted to desktop web browsers).
- Automation Test Script Execution (This test cycle focuses strictly on manual functional execution; Selenium scripts are out of scope for TP-SD-001).

4. Testing Strategy

A risk-based functional manual testing approach will be utilized for this release cycle. Critical business flows (Login -> Add to Cart -> Checkout -> Confirmation) will receive the highest execution priority.

Testing Technique	Purpose
Functional Testing	Verify each feature operates according to specified functional requirements.
Positive Testing	Verify the system processes valid business inputs seamlessly.
Negative Testing	Verify proper error handling and validation messages when invalid data is provided.
Boundary Testing	Verify input fields (e.g., text limits, numeric postal codes) at boundary thresholds.
UI & Usability Testing	Ensure visual consistency, layout alignment, typography, and intuitive user navigation.
Integration Testing	Validate end-to-end data flow between inventory, cart, and checkout modules.
Regression Testing	Confirm existing core capabilities remain unaffected after bug fixes or environment updates.
Exploratory Testing	Uncover non-obvious defects through unscripted, intuitive application exploration.

5. Test Environment

Testing will be conducted in a controlled environment across target browser baseline versions:

Component	Configuration Specification
Application	SauceDemo Web Application
Environment URL	https://www.saucedemo.com/ (Demo Environment)
Operating System	Windows 11 Professional (64-bit)
Browser 1 (Primary)	Google Chrome (v128.0 or latest stable release)
Browser 2	Mozilla Firefox (v129.0 or latest stable release)
Browser 3	Microsoft Edge (v128.0 or latest stable release)
Network Baseline	Stable Broadband Connection (20 Mbps)
Testing Methodology	Manual Test Execution

6. Test Data

Field / Flow	Test Input Data	Expected Behavior
Valid Login	standard_user / secret_sauce	Successful authentication; redirected to inventory page.
Locked Out User	locked_out_user / secret_sauce	Error displayed: 'Epic sadface: Sorry, this user has been locked out.'
Invalid Password	standard_user / invalid_pass	Error displayed: 'Epic sadface: Username and password do not match...'
Blank Credentials	[Empty] / [Empty]	Error displayed: 'Epic sadface: Username is required'
Checkout Fields	First: Test Last: User Zip: 1207	Customer info accepted; user navigates to Checkout Overview.

7. Roles and Responsibilities (Consolidated)

Role	Responsibilities
SQA Engineer	Prepare test plan, write test scenarios & test cases, execute manual test suites, log defects, perform regression testing upon fix delivery, and archive test evidence.
QA Lead / Test Lead	Review and approve test plan/cases, monitor execution progress, resolve blockers, and sign off on test completion.
Software Developer	Analyze reported defects, perform root-cause analysis, deploy fixes to test environment, and provide technical resolution notes.
Project Manager / Lead	Review overall quality metrics, evaluate project risks, and approve final release readiness.

8. Entry & Exit Criteria

8.1 Entry Criteria

- SauceDemo test environment is fully accessible and stable.
- Test Plan document is reviewed, approved, and baselined.
- Test Scenarios and Test Cases are fully documented and peer-reviewed.
- Test credentials and test datasets are verified and accessible.
- Test execution environment (Windows 11, browsers) is configured.

8.2 Exit Criteria

- 100% of planned test cases have been executed.
- 100% of Critical and High priority test cases achieve PASS status.
- No unresolved Critical or High severity defects remain open.
- All identified defects are logged with evidence and assigned for remediation.

- Final Test Execution Report and Quality Summary are published.

9. Risks, Constraints & Mitigation Strategies

Risk / Constraint	Impact	Probability	Mitigation Strategy
Environment Downtime	High	Medium	Schedule execution during stable hours; notify admin immediately upon outage.
Network Instability	Medium	Medium	Utilize fallback connection (mobile hotspot) to ensure execution continuity.
Cross-Browser Discrepancy	Medium	Medium	Execute critical smoke tests on Chrome, Firefox, and Edge early in the cycle.
Scope Creep / Automation Drift	Medium	Low	Enforce strict scope boundary (Manual system testing only for TP-SD-001).
Short Execution Timeline	High	Medium	Prioritize core business flows (Login -> Cart -> Checkout) over minor cosmetic checks.

10. Deliverables

The SQA team will deliver the following formal artifacts upon project completion:

- Software Test Plan Document (TP-SD-001)
- Test Scenarios & Executable Test Cases
- Defect Reports (Logged in Tracking Tool)
- Test Execution Summary & Sign-off Report
- Screenshots and Video Evidence Repository

PART 2: TEST CHECKLIST - DARAZ ONLINE SHOPPING

Checklist Overview & Metadata

Property	Value
Document ID	CHK-DARAZ-001
Version	1.0
Application Under Test	Daraz Bangladesh (Web Application)
Testing Type	Functional & Operational Edge-Case Testing
Testing Level	System Testing
Prepared By	Srabony Islam
Date	02.09.2026

Note: SauceDemo (used in Part 1) does not provide a registration feature, so Daraz — a live e-commerce platform — was used here to demonstrate Login, Registration, and Checkout checklist validation points as required by the assignment.

1. Login Module Checklist

ID	Validation Point	Expected Result	Status
LOG-01	Verify Login option is available	Login option should be visible and clickable	☐ Pass ☐ Fail ☐ N/A
LOG-02	Verify Login page loads correctly	Login page should load without errors	☐ Pass ☐ Fail ☐ N/A
LOG-03	Verify mobile/email field	Field should be displayed correctly	☐ Pass ☐ Fail ☐ N/A
LOG-04	Verify password field	Password field should be displayed correctly	☐ Pass ☐ Fail ☐ N/A
LOG-05	Verify password masking	Password should be hidden while entering	☐ Pass ☐ Fail ☐ N/A
LOG-06	Login with valid credentials	User should be logged in successfully	☐ Pass ☐ Fail ☐ N/A
LOG-07	Login with invalid username/mobile/email	Appropriate error message should be displayed	☐ Pass ☐ Fail ☐ N/A
LOG-08	Login with invalid password	Appropriate error message should be displayed	☐ Pass ☐ Fail ☐ N/A
LOG-09	Login with invalid username and password	Login should be rejected	☐ Pass ☐ Fail ☐ N/A

ID	Validation Point	Expected Result	Status
LOG-10	Login with blank username/mobile/email	Validation message should be displayed	☐ Pass ☐ Fail ☐ N/A
LOG-11	Login with blank password	Validation message should be displayed	☐ Pass ☐ Fail ☐ N/A
LOG-12	Login with all fields blank	Login should not proceed	☐ Pass ☐ Fail ☐ N/A
LOG-13	Verify Login button	Login button should be visible and functional	☐ Pass ☐ Fail ☐ N/A
LOG-14	Verify Forgot Password option	Password recovery option should work correctly	☐ Pass ☐ Fail ☐ N/A
LOG-15	Verify Sign Up navigation	User should be redirected to registration	☐ Pass ☐ Fail ☐ N/A
LOG-16	Verify successful login redirection	User should be redirected to the appropriate page	☐ Pass ☐ Fail ☐ N/A
LOG-17	Verify incorrect credentials cannot authenticate	User should remain logged out	☐ Pass ☐ Fail ☐ N/A
LOG-18	Verify login using registered account	Registered user should successfully log in	☐ Pass ☐ Fail ☐ N/A
LOG-19	Verify logout functionality	User should be logged out successfully	☐ Pass ☐ Fail ☐ N/A
LOG-20	Verify access to account after logout	Protected account features should require login	☐ Pass ☐ Fail ☐ N/A

2. Registration Module Checklist

ID	Validation Point	Expected Result	Status
REG-01	Verify Sign Up option is available	Sign Up option should be visible and clickable	☐ Pass ☐ Fail ☐ N/A
REG-02	Verify registration page loads	Registration page should load correctly	☐ Pass ☐ Fail ☐ N/A
REG-03	Verify mobile number field	Mobile number field should be displayed	☐ Pass ☐ Fail ☐ N/A
REG-04	Enter valid mobile number	Valid mobile number should be accepted	☐ Pass ☐ Fail ☐ N/A
REG-05	Enter invalid mobile number	Appropriate validation should be displayed	☐ Pass ☐ Fail ☐ N/A
REG-06	Submit blank mobile number	Registration should not proceed	☐ Pass ☐ Fail ☐ N/A
REG-07	Verify password field	Password field should be displayed	☐ Pass ☐ Fail ☐ N/A

ID	Validation Point	Expected Result	Status
REG-08	Verify password masking	Password should not be visible as plain text	☐ Pass ☐ Fail ☐ N/A
REG-09	Enter valid password	Valid password should be accepted	☐ Pass ☐ Fail ☐ N/A
REG-10	Enter invalid/weak password	Appropriate validation should be displayed	☐ Pass ☐ Fail ☐ N/A
REG-11	Register with already registered mobile number	Duplicate registration should be prevented	☐ Pass ☐ Fail ☐ N/A
REG-12	Verify OTP functionality where supported	OTP should be sent and verified correctly when OTP verification is required	☐ Pass ☐ Fail ☐ N/A
REG-13	Enter valid OTP	Valid OTP should be accepted	☐ Pass ☐ Fail ☐ N/A
REG-14	Enter incorrect OTP	Registration should not be completed	☐ Pass ☐ Fail ☐ N/A
REG-15	Verify OTP resend functionality	User should be able to request a new OTP where supported	☐ Pass ☐ Fail ☐ N/A
REG-16	Submit registration with missing required information	Validation messages should be displayed	☐ Pass ☐ Fail ☐ N/A
REG-17	Verify terms and conditions	Required agreement should be validated	☐ Pass ☐ Fail ☐ N/A
REG-18	Verify registration button	Registration button should work correctly	☐ Pass ☐ Fail ☐ N/A
REG-19	Verify successful registration	User account should be created successfully	☐ Pass ☐ Fail ☐ N/A
REG-20	Login using newly registered account	Newly registered user should be able to log in	☐ Pass ☐ Fail ☐ N/A

Note: Daraz may change registration fields, OTP behavior, or verification requirements based on its current implementation. During execution, unavailable functionality should be recorded as **N/A** rather than automatically marked as failed.

3. Shopping Cart & Checkout Module Checklist

ID	Validation Point	Expected Result	Status
CHK-01	Verify product can be added to cart	Selected product should be added successfully	☐ Pass ☐ Fail ☐ N/A

ID	Validation Point	Expected Result	Status
CHK-02	Verify selected product appears in cart	Correct product should be displayed	☐ Pass ☐ Fail ☐ N/A
CHK-03	Verify product name in cart	Product name should match selected product	☐ Pass ☐ Fail ☐ N/A
CHK-04	Verify product price in cart	Correct product price should be displayed	☐ Pass ☐ Fail ☐ N/A
CHK-05	Verify product quantity	Correct quantity should be displayed	☐ Pass ☐ Fail ☐ N/A
CHK-06	Increase product quantity	Quantity should increase correctly	☐ Pass ☐ Fail ☐ N/A
CHK-07	Decrease product quantity	Quantity should decrease correctly	☐ Pass ☐ Fail ☐ N/A
CHK-08	Remove product from cart	Selected product should be removed	☐ Pass ☐ Fail ☐ N/A
CHK-09	Proceed from cart to checkout	User should proceed to checkout successfully	☐ Pass ☐ Fail ☐ N/A
CHK-10	Verify delivery address section	Delivery address section should be displayed	☐ Pass ☐ Fail ☐ N/A
CHK-11	Enter valid delivery information	Valid information should be accepted	☐ Pass ☐ Fail ☐ N/A
CHK-12	Submit checkout with missing required information	Appropriate validation should be displayed	☐ Pass ☐ Fail ☐ N/A
CHK-13	Verify delivery options	Applicable delivery options should be displayed	☐ Pass ☐ Fail ☐ N/A
CHK-14	Verify delivery/shipping charge	Applicable delivery charge should be calculated correctly	☐ Pass ☐ Fail ☐ N/A
CHK-15	Verify product subtotal	Product subtotal should be calculated correctly	☐ Pass ☐ Fail ☐ N/A
CHK-16	Verify applying valid voucher/promo code where supported	Valid voucher should be applied correctly when the voucher is eligible for the selected order	☐ Pass ☐ Fail ☐ N/A
CHK-17	Apply invalid/expired voucher	Invalid voucher should be rejected appropriately	☐ Pass ☐ Fail ☐ N/A
CHK-18	Verify discount calculation	Applied discount should be reflected correctly	☐ Pass ☐ Fail ☐ N/A
CHK-19	Verify final payable amount	Final amount should be calculated correctly	☐ Pass ☐ Fail ☐ N/A
CHK-20	Verify available payment methods	Applicable payment methods should be displayed	☐ Pass ☐ Fail ☐ N/A
CHK-21	Verify Cash on Delivery where applicable	COD should be selectable when it is available for the selected order	☐ Pass ☐ Fail ☐ N/A

ID	Validation Point	Expected Result	Status
CHK-22	Verify online payment option	Selected payment method should work correctly	☐ Pass ☐ Fail ☐ N/A
CHK-23	Verify order placement button	Order placement button should function correctly	☐ Pass ☐ Fail ☐ N/A
CHK-24	Place order with valid information	Order should be placed successfully	☐ Pass ☐ Fail ☐ N/A
CHK-25	Verify order confirmation	Order confirmation should be displayed	☐ Pass ☐ Fail ☐ N/A
CHK-26	Verify order appears in order history	Successfully placed order should be recorded	☐ Pass ☐ Fail ☐ N/A
CHK-27	Verify cart after successful order	Cart should be updated appropriately	☐ Pass ☐ Fail ☐ N/A
CHK-28	Verify product availability during checkout	Unavailable products should not be incorrectly purchasable	☐ Pass ☐ Fail ☐ N/A
CHK-29	Verify checkout with multiple products	All selected products should appear correctly	☐ Pass ☐ Fail ☐ N/A
CHK-30	Verify total for multiple products	Total should reflect quantity, price, discounts and charges	☐ Pass ☐ Fail ☐ N/A

4. Test Execution Summary Dashboard

Category / Module	Total Checkpoints	Passed	Failed	Blocked	N/A	Status Result
Login Module	20	0	0	0	0	Pending Execution
Registration Module	20	0	0	0	0	Pending Execution
Checkout Module	30	0	0	0	0	Pending Execution
TOTAL SUMMARY	70	0	0	0	0	READY FOR EXECUTION